Workshop on Multinomial-Processing-Tree (MPT) Modelling

Advanced Features of multiTree

Philipps Universität Marburg

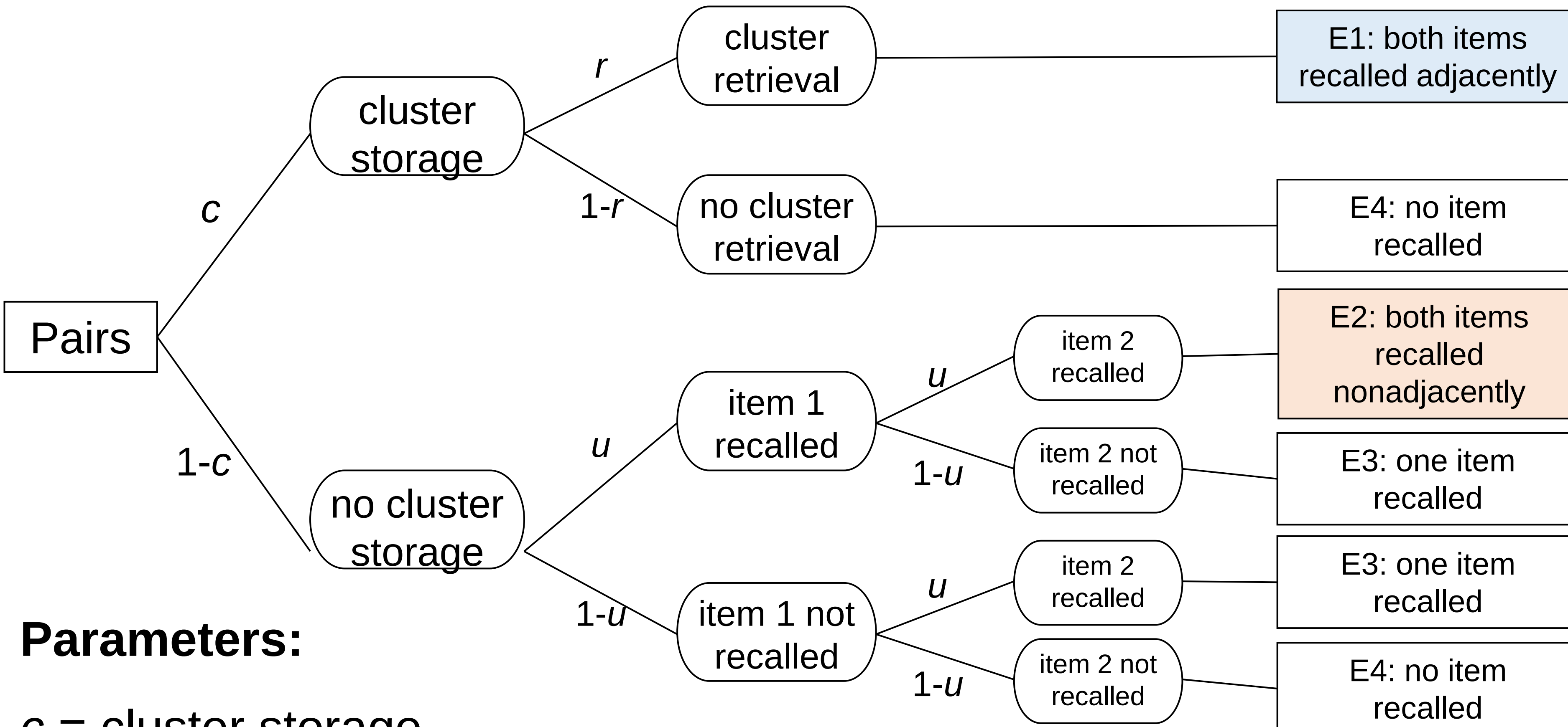
Prof. Dr. Daniel Heck

The Software multiTree

Overview

1. Practice: Pair-clustering model
2. Order constraints & interactions
3. Power analysis
4. Power optimization

Pair-Clustering Model (Batchelder & Riefer, 1980, 1986)

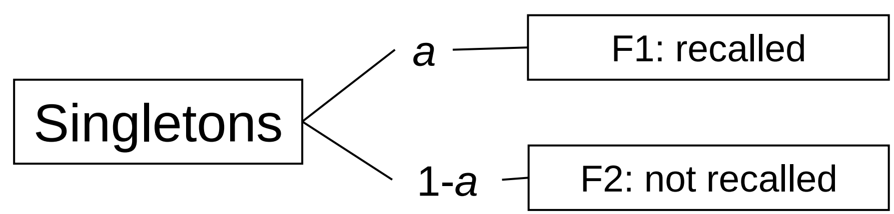


Parameters:

c = cluster storage

r = cluster retrieval

u = storage & retrieval of singletons



Practical Application

Pair-clustering model

1. Estimate the pair-clustering model for young participants (assuming u = a)
2. Estimate the pair-clustering model for young and old participants jointly
3. Does *c* differ significantly between age groups?
4. Does *r* differ significantly between age groups?

Category	Young	Old
E1	91	42
E2	14	5
E3	84	63
E4	212	290
F1	102	64
F2	298	336

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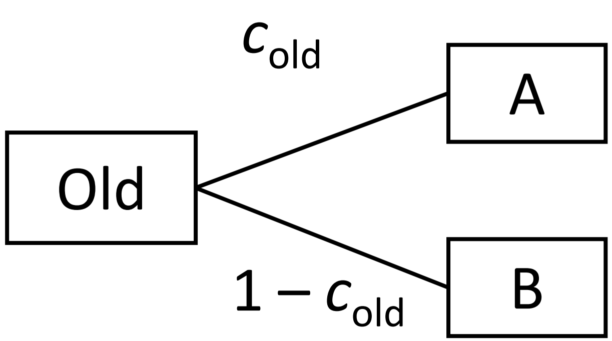
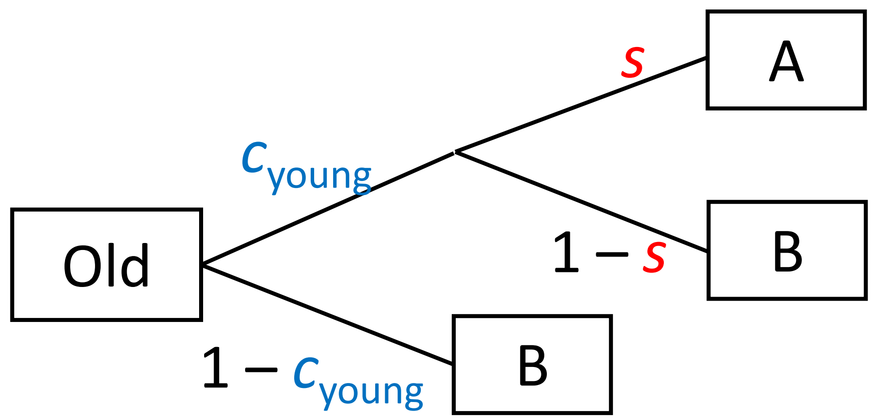
Order Constraints

- Goal: impose the order constraint $c_{\text{old}} \leq c_{\text{young}}$
- Solution: Reparameterization via $c_{\text{old}} = s \cdot c_{\text{young}}$
- We get a new parameter s = relative decline in c
  (decrease in clustering probability for old compared to young adults)

Original model:



New model:



Testing Interactions

- We want to test a hypothesis H_0 about an interaction:
- "the decline with aging is the same in storage c and retrieval *r*" (= no interaction with aging)

1. Use the reparameterization:
   - $c_{\text{old}} = s_c \cdot c_{\text{young}}$ (s_c = decline in parameter c for old vs. young)
   - $r_{\text{old}} = s_r \cdot r_{\text{young}}$ (s_r = decline in parameter r for old vs. young)

2. Test the equality constraint
   - H_0: $s_c = s_r$ (→ same decline for c and r?)

Conclusion

- There is no significant interaction effect between age (young vs. old) and memory process (cluster storage vs. retrieval) for lag 0 word pairs at least:
  - $G^2(1) = 1.14$, $p = .28$
- This is better in line with the idea of a general cognitive age decline rather than a specific decline limited to retrieval.
- Note, however, that this non-significant outcome might be due to lack of statistical power.

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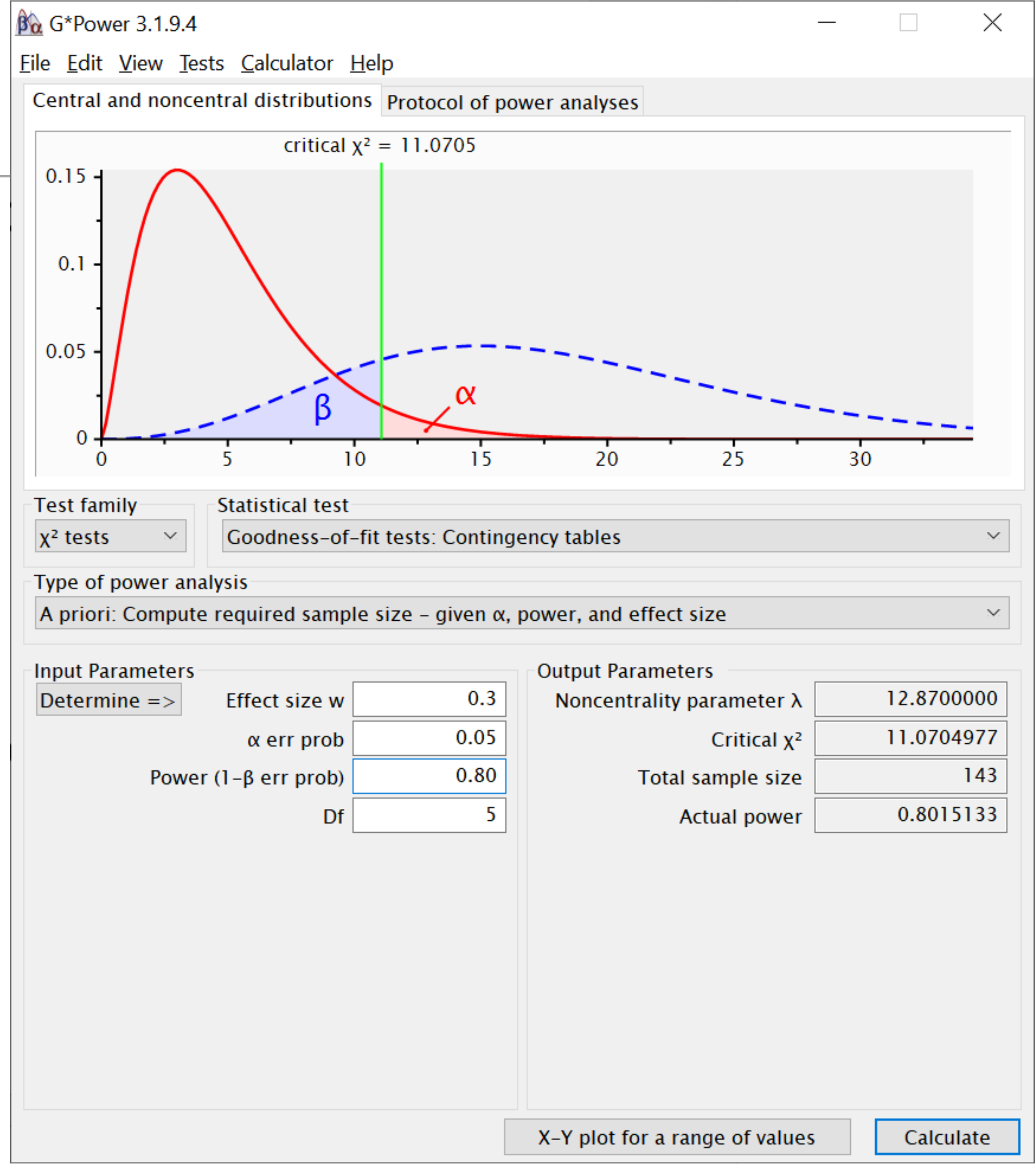
Traditional Power Analysis

- Under H_1, the G^2 statistic of H_0 follows a noncentral χ^2 distribution
  - The noncentrality parameter is: $\gamma = N \cdot w^2$
  - w denotes the effect size ($w^2 = G^2(H_0 \mid H_1$ holds given $\boldsymbol{\theta}) / N$)
- Effect size conventions
  - $w = .10$ ("small effect")
  - $w = .30$ ("medium effect")
  - $w = .50$ ("large effect")
- Types of power analysis
  - A priori: Compute required N as a function of w, α, and $1 - \beta$
  - Post hoc: Compute $1 - \beta$ as a function of w, α, and N

Traditional Power Analysis

- Example with G*Power for a "medium" effect size (w = 0.3)



Traditional Power Analysis: Limitations

- Problems:
  1. How does effect size translate into parameter values?
  2. Same meaning of effect size labels in different models?
  3. Relative size of groups/conditions is ignored.

- Cohen (1988, p. 244) on *w* effect sizes conventions:
  - "Their use requires particular caution, since, apart from their possible inaptness in a particular substantive context, what is subjectively the same degree of departure or degree of correlation (…) may yield varying *w*, and conversely. The investigator is best advised to use the conventional definitions as a general frame of reference (…) and not to take them too literally."

Approach 2: Meaningful Power Analysis

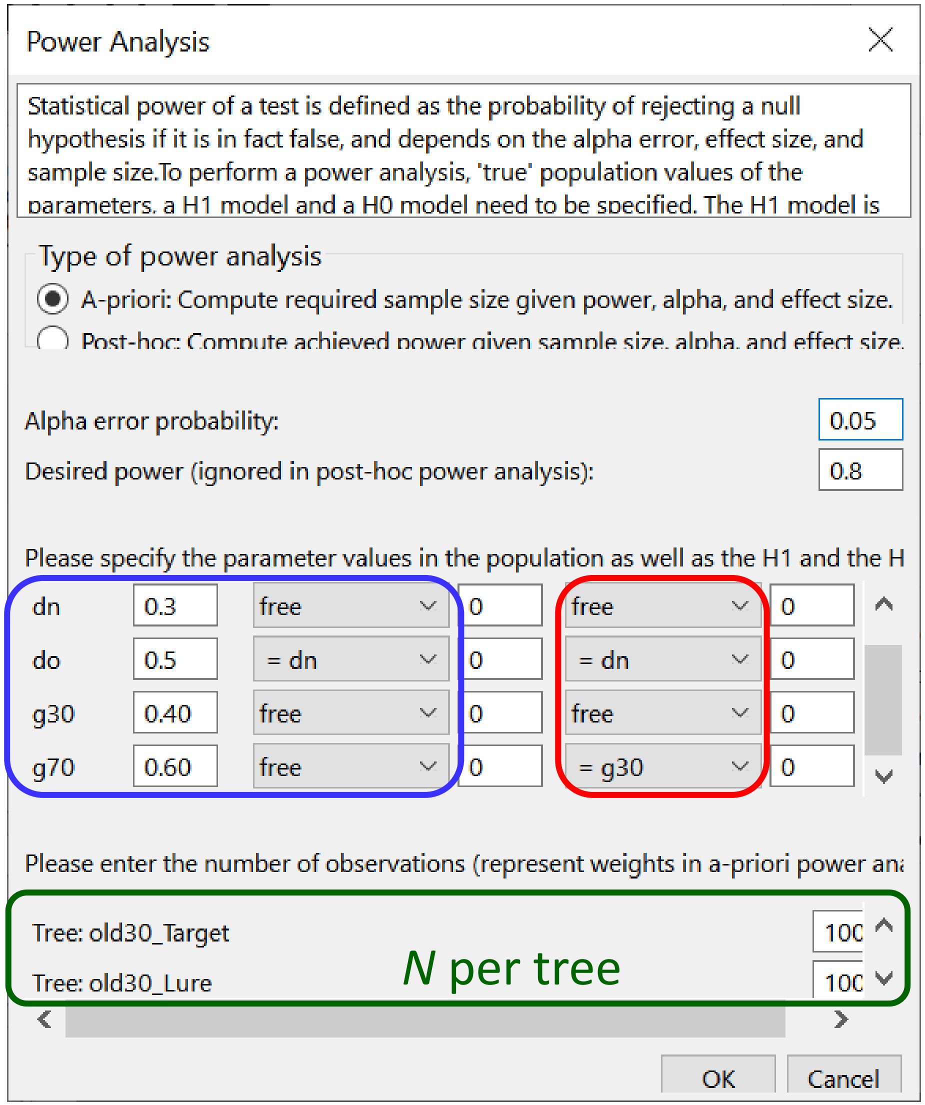
Power as a function of the model parameters θ under H_1:

1) Specify a specific H_1 model with all parameter values **θ** fixed at „plausible values"
2) Specify a nested H_0 model
3) Choose number of observations N_k for each tree k and calculate the expected frequencies under H_1
4) Fit the H_0 model to the H_1 expected frequencies by minimizing G^2
5) Use minimum G^2 value as noncentrality parameter γ
6) Compute the power $1 - \beta = P(\chi^2(\gamma, df) \geq c_{(df,\ \alpha)})$

Power Analysis in multiTree

H_1: "true state of the world"

→ power may depend on the values of *all* parameters **θ!**



H_0: nested model

→ which parameter constraints are tested

→ here: $g30 = g70$

Power Analysis in multiTree: Output

Parameters | Data | Equations | Output

```
Model specifications

              Population     H1       H0
dn        = 0.30000          free     free
do        = 0.50000          = dn     = dn
g30       = 0.40000          free     free
g70       = 0.60000          free     = g30


Required number of observations = 460.72733
Required observations per tree:
             Tree: old30_Target =  115.18183
               Tree: old30_Lure =    115.18183
            Tree: old700_Target = 115.18183
              Tree: old700_Lure =   115.18183


Model Fit
H0 Model PD^lambda=0.0 (df=2)   = 8.07460
H1 Model PD^lambda=0.0 (df=1)   = 0.22574


Effect size w                   = 0.13052
Non-centrality parameter        = 7.84886
Critical Chi-square             = 3.84146
Delta df                        = 1


Alpha error probability         = 0.05000
Requested power                 = 0.80000
Achieved power                  = 0.80000
```

Result:

→ How many observations are required?

Practical Application

Preliminaries

- Open the file: `pair-clustering model - a-priori power […].mpt`
- The file includes data and an MPT model for young and old adults
- We want to assess the power to detect a difference in recall (parameter *r*) between the two age groups.

Exercise

1. Go to „Analysis“ → „Power Analysis“ → „A-priori“
2. For the H0 Model, define: r_young0 = r_old0
3. For the H1 Model, both parameters are set to “free” with true population values: r_young0 = 0.5 and r_old0 = 0.25

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Power Optimization

Problem:

- Given a fixed total sample size and a fixed α, is there any way to maximize the power?

Answer:

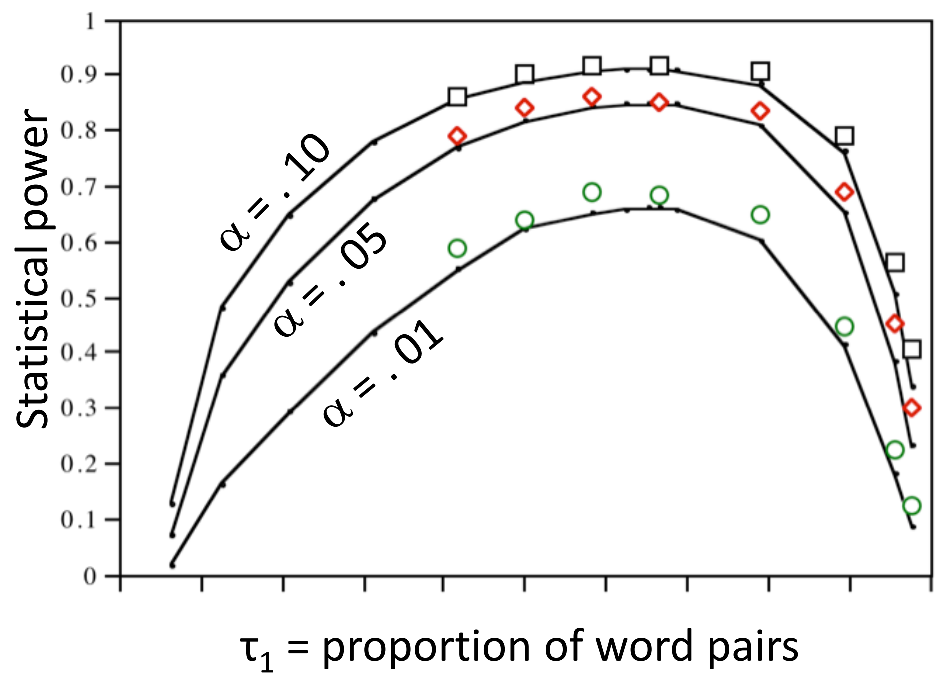
- Yes, there are many!

Optimization: Number of Observations per Tree

Example: Pair-Clustering model

- τ_1 = proportion of word pairs (relative to singletons)
- How can we optimize this proportion in the experiment?

- Plot: Statistical power of G^2 test as a function of τ_1
  - $c = r = .50$, $u = .40$, $a = .60$
  - Test of H_0: $u = a$
  - $N = 480$



- Conclusions:
  - Strong effect of τ_1 !
  - max. power for $\tau_1 = .652$
  - Thus, $\tau_1 = \ldots = \tau_K$ may be a bad default option!

Optimization: Parameter Values θ under H_1

Test-relevant parameters

- Model parameters can be divided in H_0-relevant and H_0-irrelevant parameters.
- For the pair-clustering model:
  - *u* and *a* are H_0 relevant (→ we test the H_0: $a = u$)
  - *c* and *r* are H_0 irrelevant (→ do not directly occur in hypothesis)
- Problem:
  - How to choose the values of the H_0-irrelevant parameters so as to maximize the power of the model test?

Optimization: Parameter Values $\boldsymbol{\theta}$ under H_1

- Power of the G^2 test for the pair-clustering model
  - $\alpha = .05$, $N_1 = 320$, $N_2 = 160$

Parameter values under H_1				ncp	Power
c	r	u	a	γ	$1 - \beta(\boldsymbol{\theta})$
.10	.80	.40	.60	12.49	.94
.10	.20	.40	.60	12.49	.94
.50	.80	.40	.60	8.92	.85
.50	.20	.40	.60	8.92	.85
.90	.80	.40	.60	2.53	.36
.90	.20	.40	.60	2.53	.36

Summary

- Do not ignore the power of model tests!
- G^2 is a good default option for several reasons:
  - Approximation accuracy is optimal
  - Maximum power for “diffuse” noncentrality structures
  - Option of conditional tests
- Do not forget to optimize context conditions:
  - τ_k optimization
  - θ optimization
  - conditional tests whenever possible.